

# **Al-Khwarizmi 2024 Day 1:**

## **Solutions and Notes**

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## Structure

Each problem is accompanied by

1. **Hints:** Guiding questions (ideally to be read one by one) if you are stuck while attempting the problem.
  2. **Thought processes/Motivation:** A walkthrough of how I arrived at a solution.
  3. **Solution:** A write-up in a style similar to what I would submit in a contest.
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Things to note in a solution write-up:

1. In problems that require you to find an answer (e.g. find all  $x$  such that ...), be sure to clearly state it (even if you don't have a complete proof). Ideally, state the answer at the very beginning of the write-up (though I didn't do that here for the sake of avoiding redundancy).
2. In the write-up, you can omit all thought processes and motivation—this is recommended to save time.

## 1 Problem 1

We have triangle  $ABC$  with area  $S$ . In one step we can move only one vertex at a time so that the area of the triangle during movement remains constant. Prove that we can move this triangle into any other arbitrary triangle  $DEF$  with area  $S$ .

### Hints:

- Recall the formula for the area of a triangle. How does each point move in a step?
- Based on your answer to the previous question, in what cases can you easily obtain  $\triangle DEF$  from  $\triangle ABC$ ? Consider the simplest cases, where it is possible to obtain  $\triangle DEF$  in a single step.
- Now that you know how to resolve simple cases, can you think of a way to reduce another (more complicated) case to a simple case?

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### Thought processes/Motivation:

- Before we begin attacking the problem, we must first thoroughly **digest the problem statement**. Firstly, we must understand how each point moves in a step. Recall that the area of a triangle is determined by a “base” and “height,” so if we were to move  $A$ , it means  $A$  must move parallel to  $BC$  (in order to preserve the distance between  $A$  and  $BC$ ). Similarly,  $B$  would move parallel to  $AC$  and  $C$  would move parallel to  $AB$ . Next, we should think about what it means to “move this triangle into any other arbitrary triangle  $DEF$  with area  $S$ .” One way is to view a “transformation” from  $\triangle ABC$  to  $\triangle DEF$  as “matching” each of the vertices, i.e.  $A = D, B = E, C = F$ .
- Now, we can start solving the problem by **considering simple cases**. Of course, the simplest (unresolved) case is when two pairs of vertices between  $\triangle ABC$  and  $\triangle DEF$  “match up” e.g.  $A = D$  and  $B = E$ . In that case, all that remains is to move  $C$  to  $F$ , and we would be done.
- But what if we don’t have two pairs of vertices that match up? Clearly, we can easily force at least one pair of vertices to match up, e.g. by intentionally setting  $A$  as  $D$ . Thus, by **wishful thinking**, our task

now is to determine a way to match up two pairs of vertices from this configuration (and we would be done since we know how to resolve the case where two pairs of vertices are matched up). Let's assume  $A$  matches up with  $D$ . Now, we want to match up  $B$  with  $E$ . By wishful thinking, we realise our task can be achieved if  $BE$  is parallel to  $AC$ , because then we can move  $B$  to  $E$  in a single step. Hence, if  $AC$  is parallel to  $BE$ , we are good to go. On the other hand, if  $AC$  is not parallel to  $BE$ , we can simply fix this by moving  $C$  until it is parallel to  $BE$ , assuming  $E$  does not lie on  $AB$ . (If  $E$  lies on  $AB$ , just tweak  $E$  first so it doesn't lie on  $AB$ .) Make sure you understand this argument.

- Finally, we are done, as we have figured out how to resolve the cases where one or two pairs of vertices match up.

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**Solution:**

Suppose we position  $\triangle DEF$  so that  $D$  is at point  $A$ . We are done if we can move  $B$  to  $E$  and  $C$  to  $F$  (assuming this does not already hold). Suppose also that  $E$  does not lie on  $AB$  (else, simply move  $E$  so it doesn't lie on  $AB$ ). Notice that  $B$  moves parallel to  $AC$  and  $C$  moves parallel to  $AB$  in order to preserve  $S$ . Now, to move  $\triangle ABC$  into  $\triangle DEF$ , move  $C$  so that  $AC$  is parallel to  $BE$ , then move  $B$  to  $E$ . Finally, move  $C$  to  $F$ . ■

## 2 Problem 2

For how many  $x \in \{1, 2, 3, \dots, 2024\}$  is it possible that Bekhzod summed 2024 non-negative consecutive integers, Ozod summed  $2024+x$  non-negative consecutive integers and they got the same result?

### Hints:

- How can you transform this into an equation? What is a good way to rewrite the equation?
- Observe that the integers Bekhzod and Ozod choose are relatively “free.” Thus, what do we require?
- To answer the previous question, we require some expression to be a multiple of 4048. What expression? And what does this tell us about  $x$ ? Consider its parity and exploit “free”-ness.

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### Thought processes/Motivation:

- First, it is natural to **form an equation** based on the given problem statement. Suppose the first integer in Bekhzod’s sum is  $a$ . Hence, Bekhzod’s sum is

$$a + (a + 1) + \dots + (a + 2023) = 2024a + \frac{(2023)(2024)}{2}.$$

On the other hand, if  $b$  is the first integer in Ozod’s sum, his sum is

$$b + (b + 1) + \dots + (b + 2023 + x) = (2024 + x)b + \frac{(2023 + x)(2024 + x)}{2}.$$

Equating these expressions and simplifying (and keeping in mind that we are interested in  $x$ , which motivates us to keep  $x$  on one side), we obtain

$$x^2 + 4047x + 2xb = 4048(a - b).$$

Essentially, we are now interested in values of  $x$  such that this equation has a solution.

- Now, notice that  $a$  and  $b$  are “free,” which allows the RHS to be any multiple of 4048. Thus, we simply require the LHS to be a multiple of 4048 in order for there to be a solution.

- Observe that

$$x^2 + 4047x + 2xb = x^2 - x + 2xb + 4048x$$

is a multiple of 4048 if and only if

$$x^2 - x + 2xb$$

is a multiple of 4048. Now, we rewrite the expression as

$$x^2 - x + 2xb = x(x + (2b - 1)).$$

- Notice that  $x$  and  $x + (2b - 1)$  have **different parities**. Thus, if  $x$  is even,  $x$  must be divisible by 16 (since 16 is the largest power of 2 dividing 4048, and  $x + (2b - 1)$  is odd), then we can simply choose  $b$  so that the expression is a multiple of 4048. Similarly, if  $x$  is odd, we can also choose  $b$  so that the expression is a multiple of 4048. (Here, we are exploiting the “free”-ness of  $b$ .) So,  $x$  is either a multiple of 16 or odd.

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**Solution:**

We require

$$\begin{aligned} a + (a + 1) + \cdots + (a + 2023) &= b + (b + 1) + \cdots + (b + 2023 + x) \\ \implies 2024a + \frac{(2023)(2024)}{2} &= (2024 + x)b + \frac{(2023 + x)(2024 + x)}{2} \\ \implies x(x + (2b - 1)) &= x^2 - x + 2xb = 4048(a - b - x). \end{aligned}$$

Since  $a$  and  $b$  can be freely chosen, we simply require  $x(x + (2b - 1))$  to be a multiple of 4048. Since  $x$  and  $x + (2b - 1)$  have different parities, we require  $x$  to be a multiple of 16 if  $x$  is even (since 16 is the highest power of 2 dividing 4048), then we can choose  $b$  so that the expression is a multiple of 4048. Similarly, we can do the same if  $x$  is odd. So,  $x$  is either a multiple of 16 or odd. Therefore, the answer is

$$\left\lfloor \frac{2024}{16} \right\rfloor + \left\lceil \frac{2024}{2} \right\rceil = 126 + 1012 = 1138. \quad \blacksquare$$

### 3 Problem 3

Find all  $x, y, z \in (0, \frac{1}{2})$  such that

$$\begin{cases} (3x^2 + y^2)\sqrt{1 - 4z^2} \geq z; \\ (3y^2 + z^2)\sqrt{1 - 4x^2} \geq x; \\ (3z^2 + x^2)\sqrt{1 - 4y^2} \geq y. \end{cases}$$

#### Hints:

- The inequalities are symmetric so consider any one of them for now. How can we make this inequality easier to work with?
- Another note: Often times, with symmetric inequalities like these, you can expect a solution where  $x = y = z$ . So it's a good idea to solve for the values of  $x, y, z$  in that case. (This is also a hint to the first hint.)
- To answer the first question, we can do so by somehow imposing some condition so that we can obtain an inequality in one variable. (This is, of course, easier to work with than three variables.) How can we do so? (Again, use the previous hint as a hint.)
- Adding on to the previous question, how can I, say, swap  $x$  and  $y$  (in the first given inequality) for  $z$ ?

#### Thought processes/Motivation:

- As mentioned, we can expect a solution where  $x = y = z$ . Even if that's not the case, it can still be useful to try it. (Generally, it's always good to **try simple things**, and assuming variables are equal is one way to do so.) Thus, assuming  $x = y = z$ , the given inequality yields

$$4z^2\sqrt{1 - 4z^2} \geq z \implies 0 \geq (8z^2 - 1)^2.$$

Recall that squares are always non-negative, so this forces

$$8z^2 - 1 = 0 \implies z = \frac{1}{\sqrt{8}}.$$

- This is nice because it leaves only one possibility for  $x, y, z$ . Moreover, if the inequality sign were to change from  $\geq$  to  $>$ , we would have no solution at all.
- Now, recall that previously we assumed  $x = y = z$ , which gives us an inequality in one variable that can easily be solved. How can we obtain something similar, without assuming  $x = y = z$ ? The trick is considering the **maximal variable**. WLOG, suppose  $z$  is maximal among  $x, y, z$ . (The variables are symmetric, which is why we are allowed to consider a maximal variable WLOG—this is not the case if the variables are asymmetric. In fact, considering the **order of the size of variables** in inequalities is a very common trick e.g. assume WLOG that  $x \geq y \geq z$ , but again, only do this when the variables are **symmetric**, else the argument is flawed.) Then, this allows us to transform the first inequality into

$$4z^2\sqrt{1-4z^2} \geq (3x^2 + y^2)\sqrt{1-4z^2} \geq z,$$

which is exactly the inequality as before!

- Recall also that the equality in the  $\geq$  sign is necessary for a solution. This means that we must have  $x = y = z$ , else

$$4z^2\sqrt{1-4z^2} > (3x^2 + y^2)\sqrt{1-4z^2} \geq z \implies 0 > (8z^2 - 1)^2,$$

which is clearly a contradiction. Hence, we conclude that  $x = y = z$  and so we must have  $x = y = z = \frac{1}{\sqrt{8}}$ .

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**Solution:**

WLOG, assume  $z = \max(x, y, z)$ . If  $z > x$  or  $z > y$ , then the first inequality implies that

$$4z^2\sqrt{1-4z^2} > (3x^2 + y^2)\sqrt{1-4z^2} \geq z \implies 0 > (8z^2 - 1)^2,$$

which is clearly a contradiction. Hence, we conclude that  $x = y = z$ . Replacing  $x$  and  $y$  for  $z$ , this yields

$$4z^2\sqrt{1-4z^2} \geq z \implies 0 \geq (8z^2 - 1)^2 \implies x = y = z = \frac{1}{\sqrt{8}}. \quad \blacksquare$$



## 4 Problem 4

We call a permutation of the set of real numbers  $\{a_1, \dots, a_n\}$ ,  $n \in \mathbb{N}$  “average increasing” if the arithmetic mean of its first  $k$  elements for  $k = 1, \dots, n$  form a strictly increasing sequence.

- 1) Depending on  $n$ , determine the smallest number that can be the last term of some average increasing permutation of the numbers  $\{1, \dots, n\}$ ;
- 2) Depending on  $n$ , determine the lowest position (in some general order) that the number  $n$  can be achieved in some average increasing permutation of the numbers  $\{1, \dots, n\}$ .

### Hints:

- Understand what it means for a permutation to be average increasing. How can we construct an average increasing permutation?
- Suppose we had an existing list of numbers and we add another number  $x$  to increase its average. What conditions are required for  $x$ ? This should help you solve the first part of the problem.
- For the second part of the problem, play around with some small values of  $n$ , and try to construct average increasing permutations while minimising the position of  $n$ . You might realise problems arise for terms after  $n$  in the permutation. Why? And how might this hint at a solution? How can you construct an average increasing permutation while minimising the position of  $n$ ?

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### Thought processes/Motivation:

- If a problem introduces some property (in this case, “average increasing”), it’s always a good idea to first **understand the property**. We can ask ourselves how to construct an average increasing permutation. We realise that if we have an existing list of numbers with average  $A$ , then adding  $x > A$  to the list would increase its average. Thus, an average increasing permutation is simply a permutation in which for all  $i$ , the  $i$ -th term is greater than the average of the first  $i - 1$  terms. A direct corollary is that all strictly increasing permutations are also average increasing permutations.

- Now, we can start with the first part of the problem. If  $x$  is the last term, then by our previous argument, we simply require

$$x > \frac{(1 + 2 + \cdots + n) - x}{n - 1}.$$

Solving this, we obtain

$$x > \frac{\frac{n(n+1)}{2} - x}{n - 1} \implies x > \frac{n + 1}{2} \implies x \geq \lfloor \frac{n + 1}{2} \rfloor + 1.$$

Indeed, the permutation

$$1, 2, \dots, n, \lfloor \frac{n + 1}{2} \rfloor + 1$$

works. (We know to arrange the remaining terms in ascending order since we previously realised that strictly increasing permutations are also average increasing permutations.)

- Next, we move on to the second part of the problem. It is useful to **experiment with small values** of  $n$ . We eventually realise that in a permutation, problems arise after the term  $n$  because  $n$  is relatively large, meaning it pulls up the average, but recall that we require all terms after  $n$  to be strictly greater than the average of the terms up to  $n$ . Suppose in an average increasing permutation where the position of  $n$  is minimised, we have that  $x$  is the minimal term after  $n$ . This means that the minimal average of terms up to  $n$  is  $\frac{1+2+\cdots+(x-1)+n}{x}$ . Thus,

$$\begin{aligned} x &> \frac{1 + 2 + \cdots + (x - 1) + n}{x} \\ &\implies x^2 + x > n \\ &\implies (2x + 1)^2 = 4x^2 + 4x + 1 > 8n + 1 \\ &\implies x \geq \lfloor \frac{\sqrt{8n + 1} - 1}{2} \rfloor + 1. \end{aligned}$$

Indeed, the permutation

$$1, 2, \dots, \lfloor \frac{\sqrt{8n + 1} - 1}{2} \rfloor, n, \lfloor \frac{\sqrt{8n + 1} - 1}{2} \rfloor + 1, \dots, n - 1$$

works. (Again, we know to arrange the remaining terms in ascending order by previous intuition.) We know that the permutation up to  $n$  must work. To check that the permutation works for terms after  $n$ ,

simply notice that if  $k$  is added to form an average increasing sequence, then  $k + 1$  can be added next to form an average increasing sequence because the average of the terms up to  $k$  is less than  $k$ . We realise this also explains why strictly increasing permutations work.

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**Solution:**

Notice that an average increasing permutation is simply a permutation in which for all  $1 \leq i \leq n$ , the  $i$ -th term is greater than the average of the first  $i-1$  terms. It follows that if  $k$  is a term of an average increasing sequence, the next term can (but not necessarily) be any number  $\geq k + 1$  since the average up to the term  $k$  is less than  $k$ .

1) If  $x$  is the last term, then by our previous argument, we require

$$x > \frac{(1 + 2 + \dots + n) - x}{n - 1}$$

$$\implies x > \frac{\frac{n(n+1)}{2} - x}{n - 1} \implies x > \frac{n + 1}{2} \implies x \geq \lfloor \frac{n + 1}{2} \rfloor + 1.$$

Indeed, the permutation

$$1, 2, \dots, n, \lfloor \frac{n + 1}{2} \rfloor + 1$$

works by our previous argument, so the answer is  $\lfloor \frac{n+1}{2} \rfloor + 1$ .

2) Suppose in an average increasing permutation where the position of  $n$  is minimised, we know that  $x$  is the minimal term after  $n$ . This means that the minimal average of terms up to  $n$  is  $\frac{1+2+\dots+(x-1)+n}{x}$ . Thus,

$$x > \frac{1 + 2 + \dots + (x - 1) + n}{x}$$

$$\implies x^2 + x > n \implies (2x+1)^2 = 4x^2 + 4x + 1 > 8n + 1 \implies x \geq \lfloor \frac{\sqrt{8n+1}-1}{2} \rfloor + 1.$$

Indeed, the permutation

$$1, 2, \dots, \lfloor \frac{\sqrt{8n+1}-1}{2} \rfloor, n, \lfloor \frac{\sqrt{8n+1}-1}{2} \rfloor + 1, \dots, n - 1$$

works by our previous argument, so the answer is the  $(\lfloor \frac{\sqrt{8n+1}-1}{2} \rfloor + 1)$ -th position. ■